Lecture Transcript by Slide

Week 3 · Chapter 2: Waves and Measurements · 33 min (first part of a 1 h 44 min recording)
Auto-transcribed, so expect occasional errors, especially in technical terms. Timestamps are positions in the recording.

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0:01:06 Rule of Electrical Flow: Positive and Negative Electrodes

0:01:47 Electrode Placement for Monitoring Lead II

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0:25:47 Refractory Periods

0:26:00 Each Cardiac Cycle Should Produce a Pulse

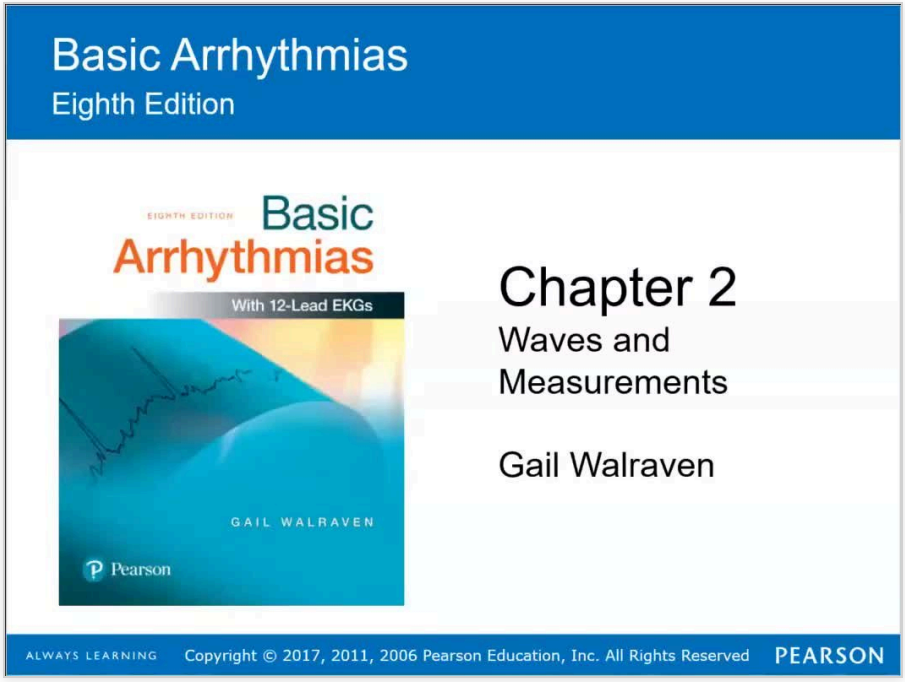
0:31:47 Various QRS Configurations

0:32:48 Artifact: Muscle Tremors

0:32:52 Artifact: Patient Movement

1 Chapter 2: Waves and Measurements (title slide)

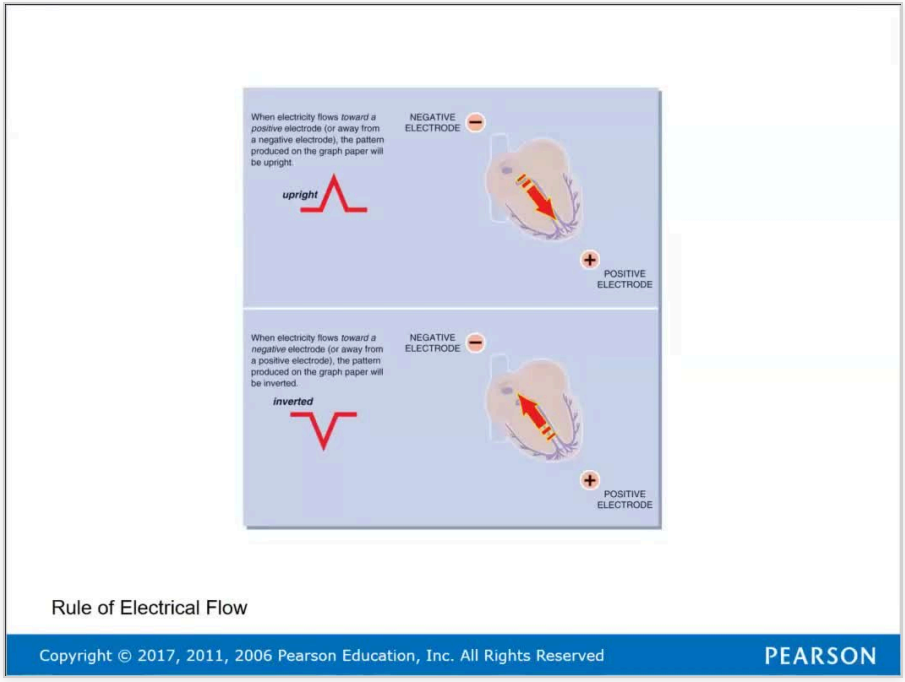
0:00:00 – 0:00:21



0:00:09 Okay, so as a quick review, because there's information in chapter two here that is important.

2 Rule of Electrical Flow (diagram)

0:00:21 – 0:01:06

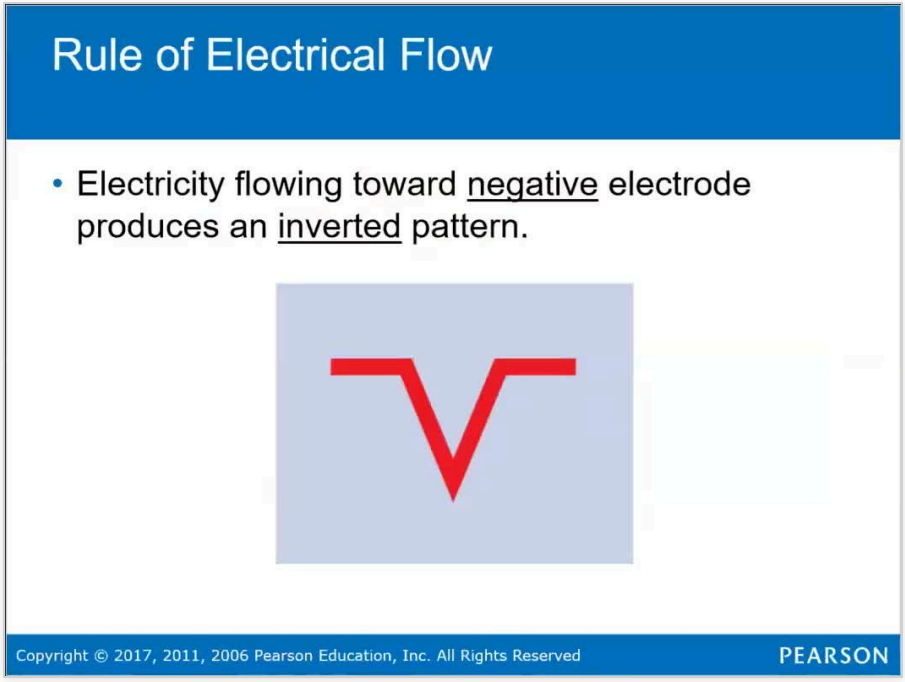


0:00:22 So we talked about how electrical flow going towards a positive electrode will create a positive deflection. And then electrical flow going towards a negative electrode, which is typically

0:00:41 the right arm is gonna give us a negative deflection in terms of the waveform, okay? How it deflects. And remember, the middle line going through the middle of the EKG is what we call the isoelectric line. It's neutral. Anything above that is positive. Anything below that is negative. Talked about that last week.

3 Rule of Electrical Flow: Positive and Negative Electrodes

0:01:06 – 0:01:47

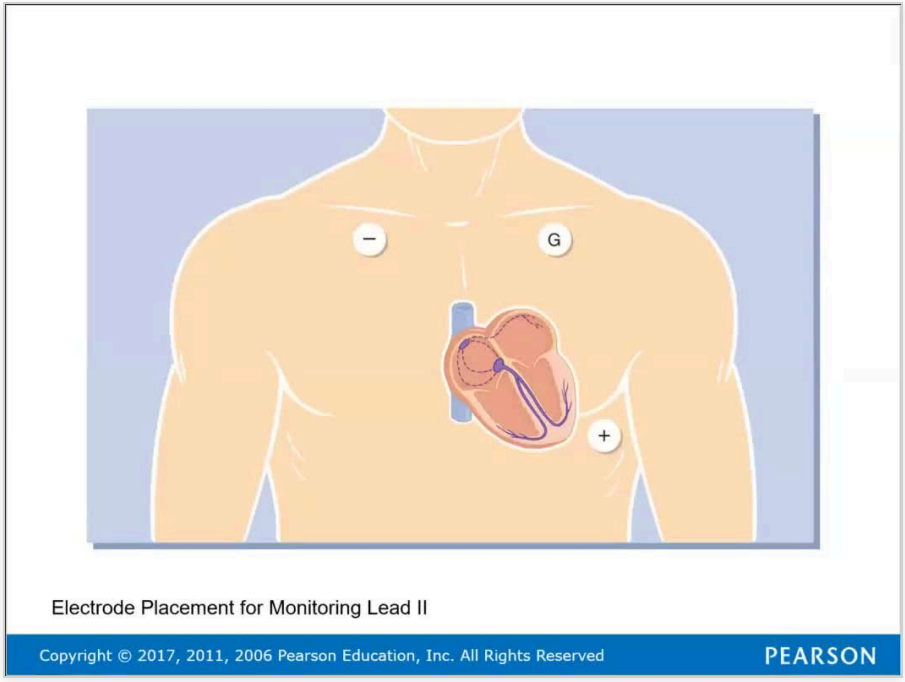


0:01:12 and sometimes when we get a negative inflection, especially when it comes to the P-wave, sometimes people will say a negatively deflected P-wave, but we also use the term inverted, that the P-wave is upside down.

0:01:31 We talked about those, I'm going to review those again today, but we talked about those last week, right? What does the P-wave represent? Oh, the P-wave. P-wave. Okay. What does the P-wave represent? Yeah. Yeah, you got it. All right. We talked about how the right arm is negative. The left,

4 Electrode Placement for Monitoring Lead II

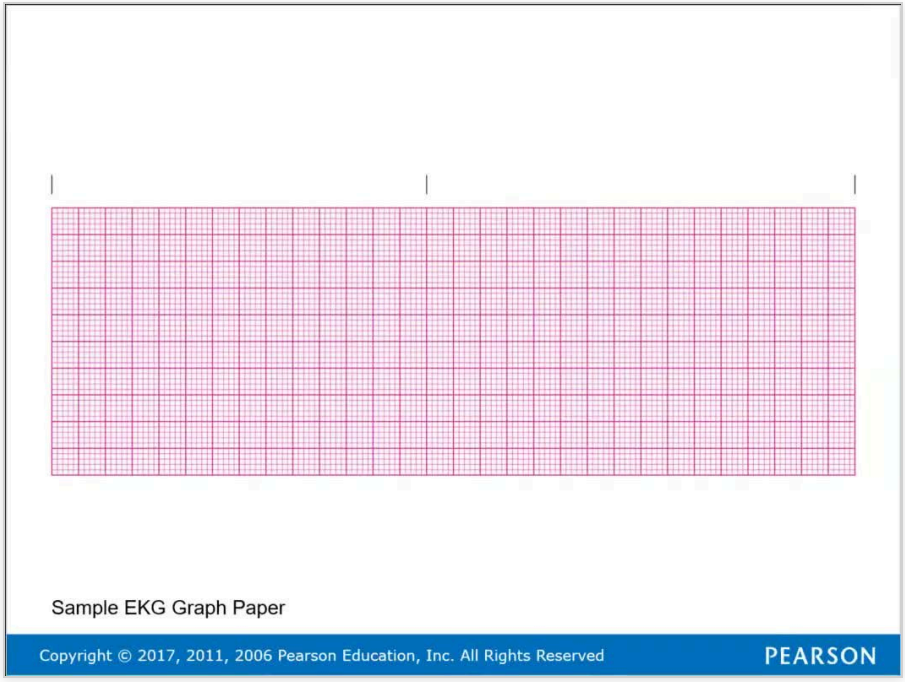
0:01:47 – 0:02:17



0:01:52 the right shoulder is negative or right arm or right shoulders as a negative. The left shoulder is ground, but it's actually a positive electrode. So when we get flow going towards that left shoulder, we're going to get a positive deflection. And then the left leg is positive as well.

5 Sample EKG Graph Paper

0:02:17 – 0:06:24



0:02:17 We talked about standard EKG paper, how the three hash marks going across the top are from one hash mark to the next is three seconds apart. To the next hash mark is three seconds apart. So when you see three hash marks on the top, that means we have a, how many seconds strip?

0:02:39 Six seconds strip. And then we talked about the large boxes, how the large boxes, which is the bold pink line, that is how much time?

0:02:57 0.2 seconds, and these tiny little boxes inside are worth how much time? 0.04, okay? So five small boxes to make up one large box,

0:03:11 five large boxes to a second, right? And that means I've got 15 small boxes to a second. Guys, you're with me on that math, right?

0:03:28 How many large boxes in a second? Each large box is 0.2 seconds. You got it? Say it. Large boxes. How many large boxes make up one second? But it would be 15 small boxes that make up one second.

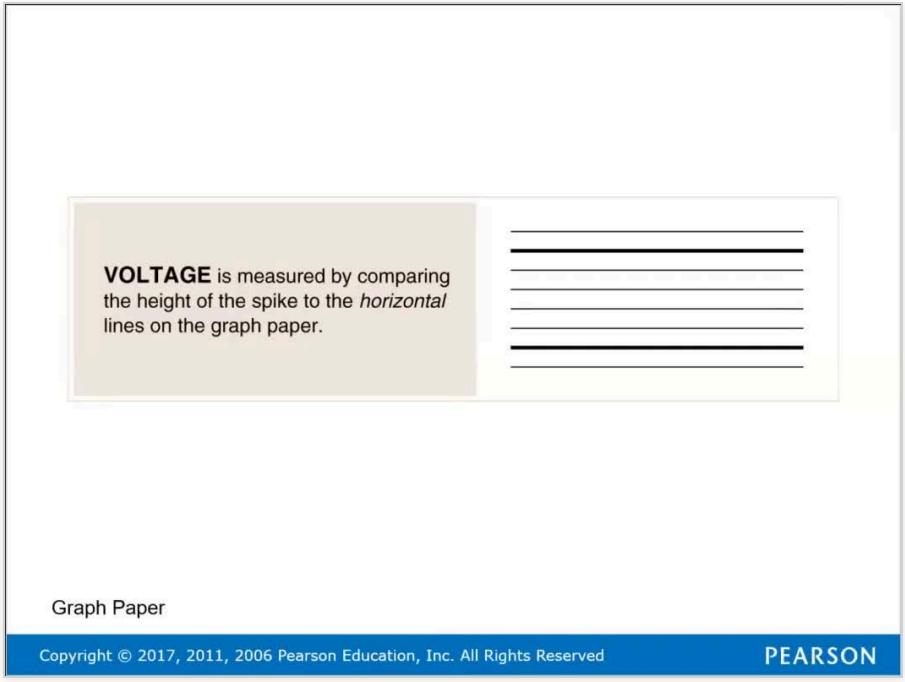
0:03:56 you're looking at me like I'm speaking a different language so one large box right which is this right here one large box is point two seconds so how many large boxes make up one second well if one small box is point two point two plus another large box is 0.4, plus another large box is 0.6, plus another large box is 0.8, plus one more makes up one, one second, right? And then if I go from hash mark to hash mark, there's going to be 15 large boxes that make up three seconds. And if there's five small boxes in each large box, right? Then how many small boxes

make up a second? That would be how many? So if I have five large boxes that make a second, so from here to here is one second. So then five times five would be 25 small boxes make up one second. Okay. Well, maybe I need to add some questions on the hourly exam. How many boxes make up a minute? Would you be in

0:05:31 trouble if I asked that question? By the way, I let you use a calculator for the test. I can extend my sarcasm and say Okay, you're allowed to take your shoes off if you need to count 123. Sorry. Sometimes I can't help myself. I'm just a little sarcasm. Okay. All right, questions on this. This is going to be important because we are going to be counting the small boxes all semester long. Okay. And by the end of the semester, I'm probably more likely to be worried about your eyes than your ability to count because if you stare at these boxes for too long. Another sarcasm you guys didn't get it. Sorry, don't worry about it. Okay, so we talked about

6 Graph Paper: Voltage

0:06:24 – 0:07:23

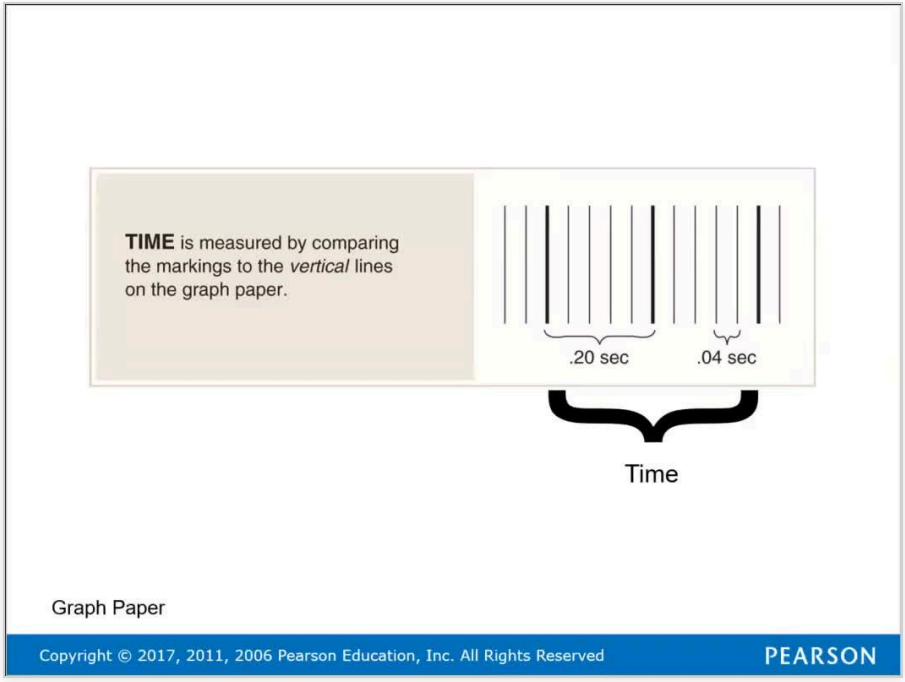


0:06:27 the horizontal lines when the waveforms go above and below the horizontal lines we're measuring voltage, right? So when the atria depolarize the height of the P wave above the amount of voltage it took for the atria to contract. Not going to be making you measure voltage, so I'm just explaining what the horizontal lines are. When the QRS complex, which represents what?

0:07:05 Ventricular contraction, right? When we see the negative Q wave followed by a positive R wave followed by a negative S wave, we can actually measure the voltage based on how far it goes above and below the isoelectric line. Okay. The vertical lines we already know are the time measurements. See that large box, 0.2 seconds, small box, 0.04 seconds. Okay.

7 Graph Paper: Time

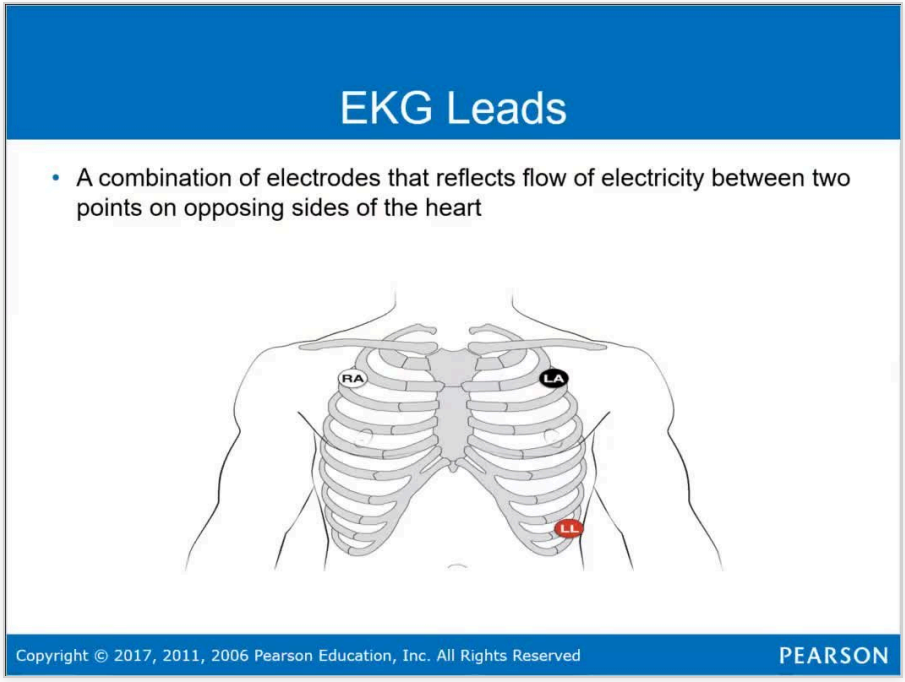
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8 EKG Leads

0:07:35 – 0:08:00

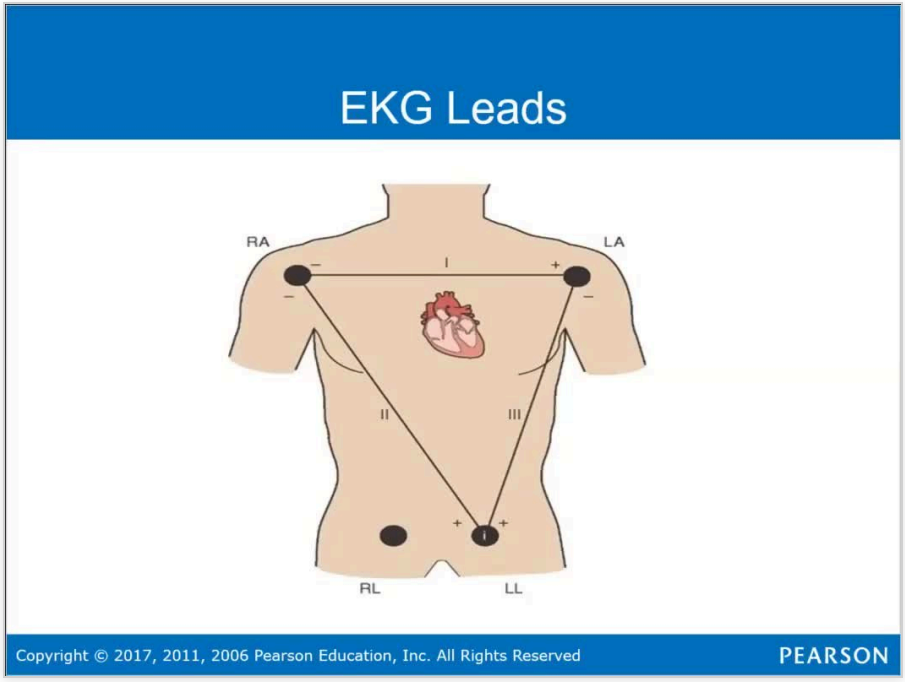


0:07:37 We talked quickly about what a lead is. A lead is a combination of two points, generally on opposing sides of the heart, that we can measure electrical activity across that lead.

0:07:56 And we talked about how when we put three leads on a patient's chest,

9 EKG Leads (diagram)

0:08:00 – 0:08:53

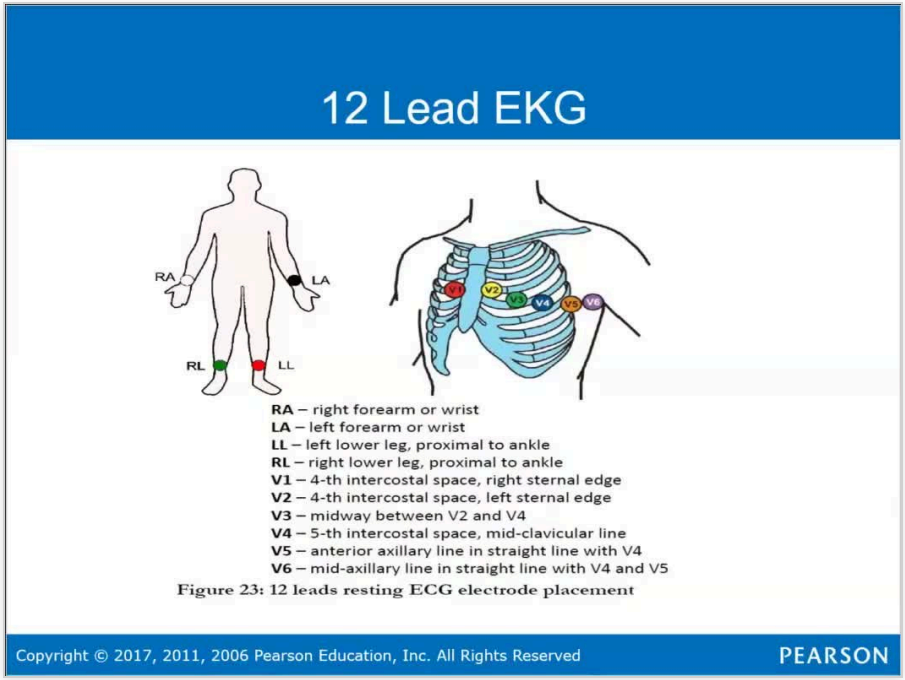


0:08:00 we create what's called Eindhoven's triangle. And then we actually put leads to these, or numbers to these leads. So right arm to left arm, which is Roman numeral one or lead one,

0:08:18 right arm to left leg, which is lead two, and then left arm to left leg, which is lead three. Okay. And you can see the electrodes are positive based on where the electrical flow is going. Anything that flows back to that right arm is going to have a negative deflection. Anything flowing towards the left leg is going to have a positive deflection. Anything flowing towards the left arm is going to have a positive deflection, okay?

10 12 Lead EKG: Electrode Placement

0:08:53 – 0:14:00



0:08:54 We talked about where do the leads go for a 12 lead EKG. We talked about how typically we try to put the electrodes on the wrists and the ankles. Those are the four limb leads. Right arm, left arm, right leg, left leg. And then we talked about putting the pre-cordial leads on the chest. And it's gonna be important for the placement, especially for those of you that are doing clinical, right? because you're going to be putting these leads on the patient when you go to clinical and performing 12 lead EKGs. So V1 goes to the right side of the sternum and the fourth intercostal space. Intercostal space is the space between ribs. So you actually are counting down to the fourth intercostal space and putting the electrode for the 12 lead EKG to the right side of the sternum. V2 goes to the fourth intercostal space, left side of the sternum.

0:09:58 Well, I mean, sometimes you can just put your fingers on there. You might have to press a little bit more firmly to figure out where that location is. All right. And then if you're, you know, you're when you go to clinical for this course, you're going to be paired up with a mentor who's going to be monitoring your electrode placement, and then help guiding you where they go. All right. That's a good question. And last week, we talked about what if somebody had an open heart surgery, where they crack open the sternum, but they have this huge bandage right here, you got to move your electrodes out a little bit, right? Or what if somebody has, you know, a right leg amputation, right? You can't put that electrode down there, right? So we talked about how we might have to vary the electrode placement, but as long as everybody's putting the electrode in the same place, right, then we should get some consistency. Because one of the things about 12 leads is when you come in and a patient

0:11:06 comes in and you're doing a 12 lead, we want to see what the electrical activity looks like in 12 different leads, right? And then what we'll do is a lot of times they'll do an EKG the next day and the next day and the next day. So they're tracking the EKG changes, they're comparing today's EKG to yesterday's EKG, okay? So as long as the electrode placement is somewhat similar, You should be able to recognize whether there's any electrical changes in the patient's cardiac conduction system separate from, oh my gosh, that lead is in the wrong spot. And the cardiologists that read this usually are pretty good at saying, okay, that's bad lead placement.

0:11:59 The next electrode we talked about placing was V4, not V3, V4, and V4 is going to go to the fourth intercostal space right at the right sternal edge. So it's going to go right down here. And then we're going to take V3 and we're going to go halfway in between V2 and V4. So it's probably going to get placed right there. And then V5, anterior axillary. We said axillary is armpit. So right at the front of the armpit, we're gonna go down to the fifth intercostal space. And then V6, mid axillary. So that's the middle of the armpit, down in the fifth intercostal space, okay?

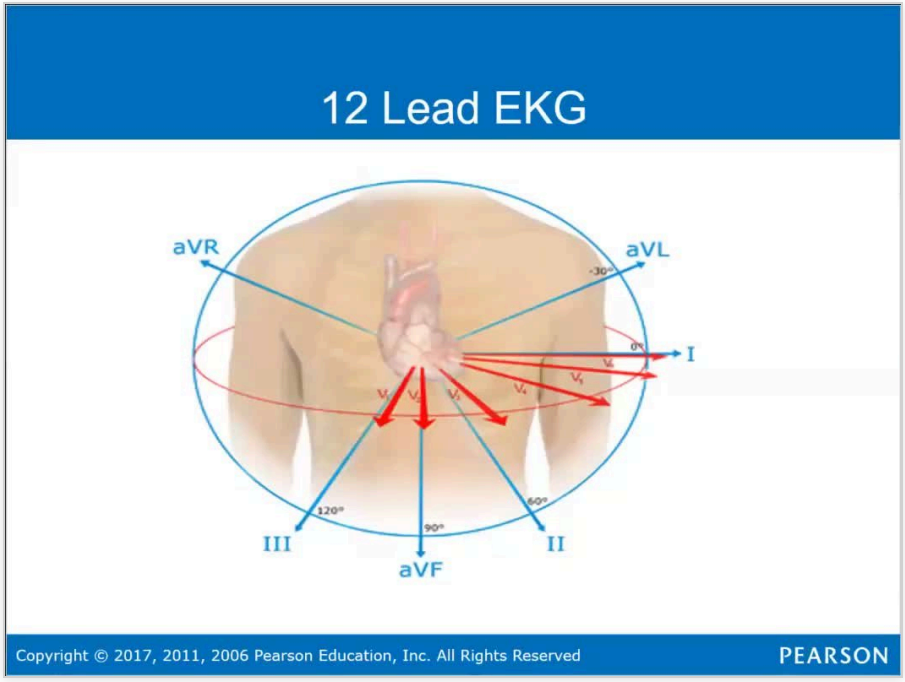
0:12:52 And those are the 10 leads that we're gonna put on to get 12 different electrical views of the patient's heart. Okay, one of the other things I talked about is the limb leads. And I talked about how we get 12 different electrical views.

0:13:12 So when we put these 10 leads on, it's going to create what's called a central terminal. So the leads can create a central terminal right in the middle of the heart. and then we can get a central terminal to the right arm, central terminal to the left arm, central terminal to the left leg, AVR, AVL, AVF.

0:13:35 And so the limb leads and the augmented leads are gonna cover the frontal plane of the heart, the electrical activity of the frontal plane, where the pre-cordial leads that V1, V2, V3, V4, V5 and V6 are gonna cover the horizontal plane of the heart, okay? The electrical activity in those frontal and horizontal planes.

11 12 Lead EKG: Lead Axes

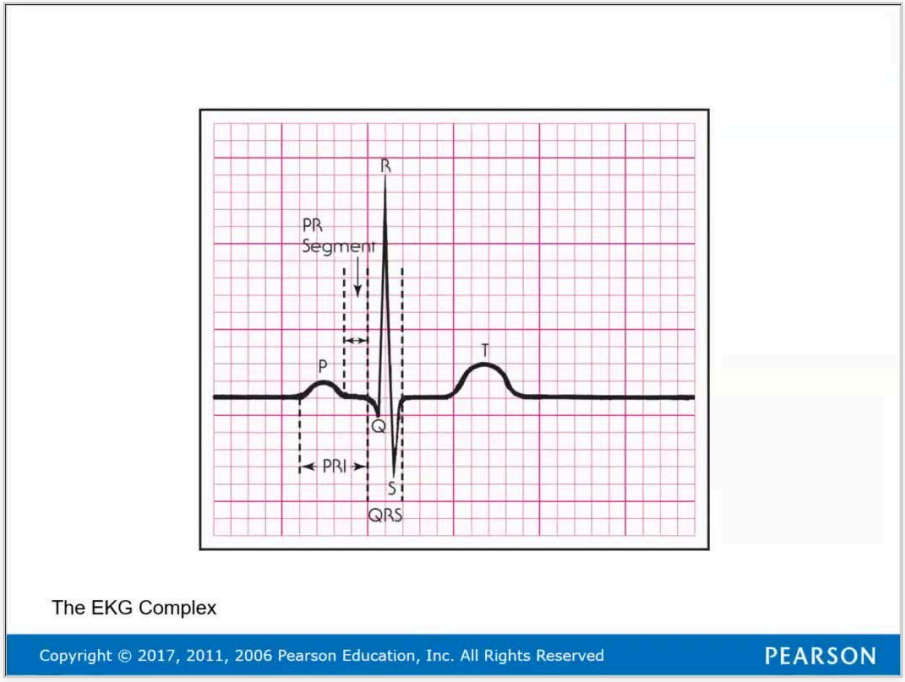
0:14:00 – 0:14:17



0:14:01 And that's what you're seeing here. You see that the red arrows are the horizontal plane, and then the blue arrows are the frontal plane, okay? Or the vertical, straight up and down.

12 The EKG Complex

0:14:17 – 0:20:53



0:14:19 We talked about this, we talked about how the P wave is the first waveform that we see it represents atrial depolarization. What's another word for depolarization?

0:14:36 Contraction. Good. We talked about the heart's electrical pathway. What's the primary pacemaker of the heart? SA node. So when the SA node fires, the impulse goes where? Over two pathways. What of the two pathways, the nodal pathway, the intranodal pathway, which is going to send it to the AV node, where else does it send the impulse? Intra atrial pathway. Don't be shy. Somebody's saying it correctly over there. And it delivers those impulses to the atrial muscle cells for contraction. We talked about how the AV node patches the impulse from the essay note and holds onto it for a short period of time. We talked about why.

0:15:27 We talked about how the heart is meant to beat from the top down. When the atria contract before the ventricles, we maximize the amount of blood in the ventricles prior to ventricular contraction. And that maximizes our stroke volume. We talked about cardiac output as a whole. What's the formula for cardiac output?

0:15:50 Yeah, we did. We talked about this briefly. you should know the formula for cardiac output. As sweet equals to, no, heart rate or volume. Stroke volume times heart rate. So the stroke volume is the amount of blood that gets ejected each time the ventricle contracts. So what's the total amount of blood each time it contracts?

0:16:23 Times the number of contractions per minute is the amount of blood that your heart beats in the minutes time. we call that cardiac output. So one of the best ways to make sure we have adequate cardiac output is to make sure that we maximize our stroke volume. Okay, so that's why it's important that the heart beats from the top down. It maximizes the amount of blood that gets pumped out to the body. So the AV node catches that impulse and it holds on to it for a short delay, allowing that contraction to occur. And then when that contraction occurs, then it sends that signal down through, what's the next step?

0:17:13 Bundle of his to the right and left bundle branches to the perkinje fibers and delivers that impulse to the ventricular muscle cells for ventricular contraction to take place. Now, when we look at this, when we record that electrical activity on paper, we have our atrial depolarization, which is represented by the P wave. And then we have this very short delay here from the end of the P wave to the beginning of the Q wave, which we call the PR segment. That's the delay at the AV node.

0:17:52 And then we talked about, and I told you I'm not going to make you measure the PR segment, but I told you I am gonna make you measure the PRI, PR interval, which encompasses both the atrial depolarization and the pause of the AV node. So you see this PR interval goes from there to there. And I gave you an example because this P wave starts right at this line and it ends right at that line. So I told you we were gonna be counting small boxes. Here we go. One, two, three, four. Small box, so our PRI is four small boxes or 0.16 seconds. Oh, here we go with the math again, right? Now, this can be potentially a strain on your eyes, so one of the things that you can do when we start looking at this is what I can do is I can take a blank piece of paper and I could make a mark on my paper where the key wave starts and where the cue wave starts. What

0:19:07 I do is bring this down to the corner here. Now I can see clearly that a storage vault boxes. I'll show you guys on this side. Did you guys see what I did? Everybody's good with that. Okay. Okay. So then we're looking at a negative cue wave followed by a positive R, followed by a negative S, and then back to the isoelectric line. So we called this QRS, we called it a complex. And what does it represent?

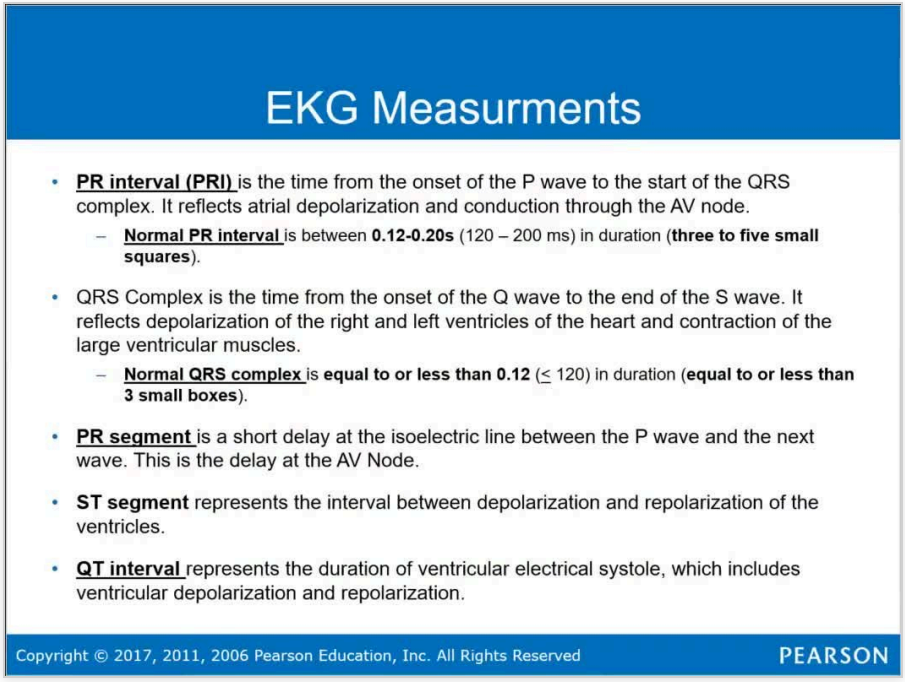
0:19:43 It represents ventricular contraction. And we are going to be measuring the width of the QRS complex as well. So again, you know, you can see that it starts right on the line and it ends on a line. And so the width of that QRS complex is two small boxes. But if you needed to, again, you could put a blank piece of paper there,

0:20:14 mark it where it starts, mark it where it finishes, and then you can see that that is two small boxes, or 0.08 seconds in time.

0:20:28 And then we have another pause at the isoelectric line, and then we finished with the T wave, and the T wave represents ventricular repolarization. So that right there is one complete heartbeat. Okay, as I said last week, we don't see atrial repolarization, only ventricular repolarization. All right, we talked about this, the PR interval

13 EKG Measurements: Definitions

0:20:53 – 0:22:32



0:20:57 from the beginning of the P wave to the beginning of the QRS complex. It represents atrial depolarization and the delay at the AV node. Normal PRI is 0.12 to 0.2 seconds. Or I could say three to five small boxes.

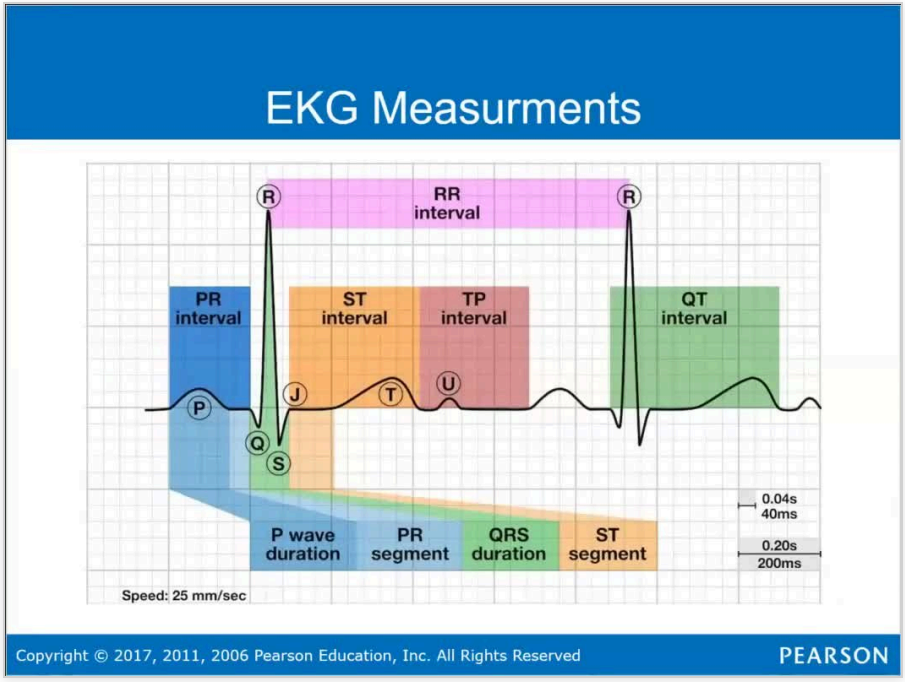
0:21:22 URS complex is from the beginning of the Q wave to the end of the S wave, and normal width is 0.12 seconds or less.

0:21:36 or I could say three equal to or less than three small boxes. The PR segment is that delay. I'm not going to make you measure that. The ST segment is from the S wave to the end of the T wave. That typically covers ventricular depolarization and repolarization. And then the QT interval represents the whole thing. There won't be any measurements there, but 12 lead EKG, when you do one, will measure all of these pieces for you electronically. When it records it, it will measure it for you.

0:22:21 And there is normals for the QT interval and the PR segment that the machine will compare the values to.

14 EKG Measurements: Intervals and Segments

0:22:32 – 0:25:47



0:22:33 Okay. And this is just graphic representation of PR interval. So from the beginning of the P wave to the beginning of the Q wave. PR segment from the end of the P wave to the beginning of the Q wave.

0:22:50 QRS complex from the beginning of the Q to the end of the S. ST segment is going to be from the end of the S wave to the end of the T wave. Now the ST segment is important, not necessarily for a measurement, but this is what a normal ST wave should look like. So it should come back to the isoelectric line, we should have a period of inactivity, and then we see ventricular repolarization. When somebody comes in the emergency department complaining of chest pain, having a heart attack, what's the official name for a heart attack?

0:23:30 Myocardial infarction, right? So when somebody's having an active MI, they're gonna have changes in this ST segment. Sometimes this S wave, so sometimes this S wave, will not come back down to the isoelectric line. It might stop right here and it might go straight across and then we hit a high T wave. That would be known as ST elevation. that is the most common finding that we see on a 12 lead for somebody that's having a heart attack when we do a 12 lead EKG. So ST elevation, by the way, the biggest reason to do a 12 lead EKG is to check for somebody having a myocardial infarction or a heart attack.

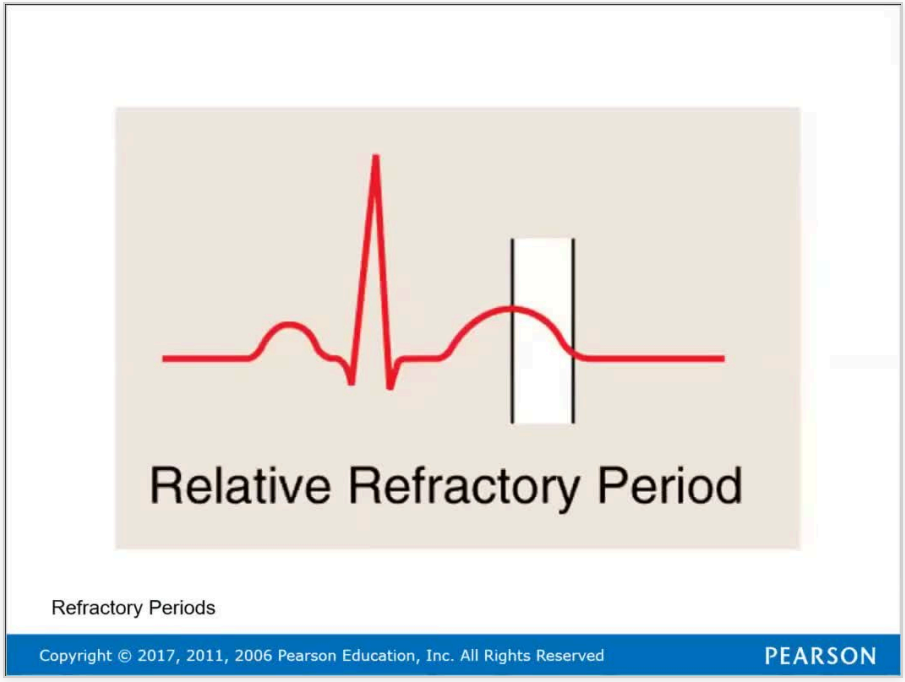
0:24:21 There's other indications to do that, but probably the main indication. and if they don't want to be coming in from the ED, you could be a patient in the hospital and all of a sudden your chest starts to feel heavy and I got jaw pain, back pain, radiating, arm pain, right? And I'm suddenly

feeling short of breath or a little bit faint, right? Probably one of the things you wanna do is put them on oxygen, get a 12-week AG and then draw some blood tests because there's a specific protein we look for that that's called troponin. Nothing you need to know for this test for this class, but that's, that's what that is. And then the other thing that I briefly mentioned last week is we're going to be counting the number of small boxes between our waves. One of the things we're going to talk about today is our systematic approach of gathering information. And one of the most important things we want to know is, is the rhythm

0:25:23 regular meaning does the heart fire at regular intervals so that the spacing between beats is the same all the way across the strip or the spacing between the beats is not the same all the way across the strip okay so we're going to be counting the number of small boxes in between these R waves across the strip it'll make a little bit more sense when we go through this and we talked

15 Refractory Periods

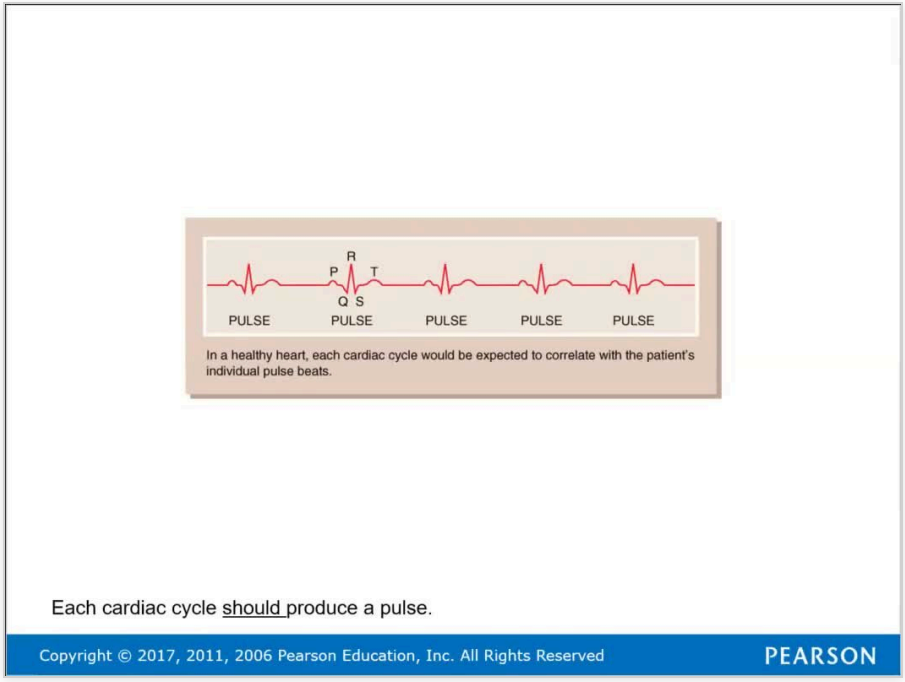
0:25:47 – 0:26:00



0:25:49 about the refractory period and the both the relative refractory period and the absolute refractory period. Okay, so I gotta actually, I apologize, but I gotta take this call just

16 Each Cardiac Cycle Should Produce a Pulse

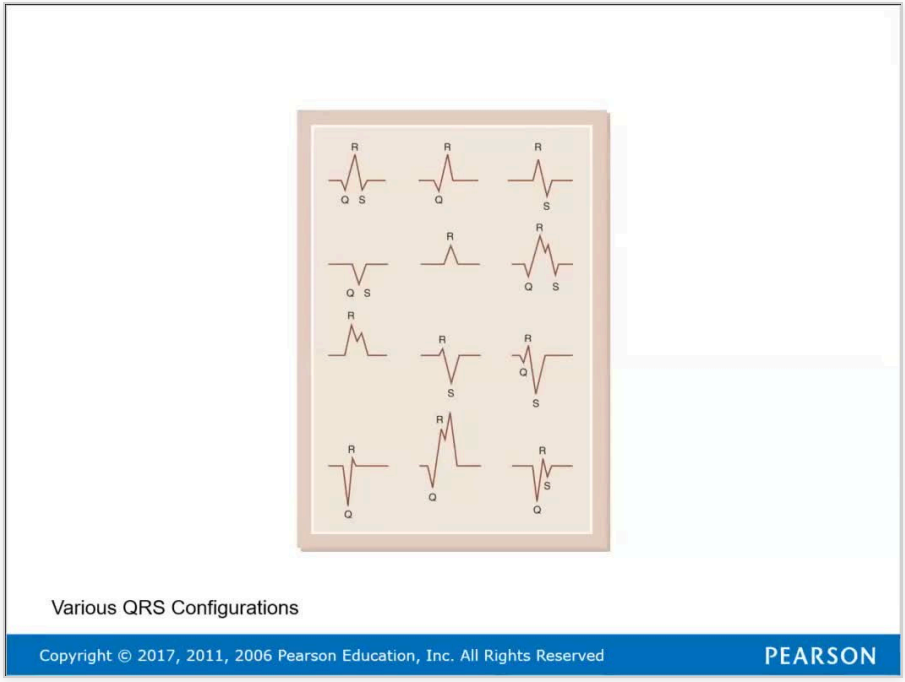
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0:26:11 two minute break. Hello. ... ... I don't know what you're saying, but I'm going to keep up with my entire thinking. Thank you so much. I think it was never good. Things didn't work, and you can't go into it. I'm not going to tell you a lot. Because everything was a good one. It was a good one. But thank you for giving me this. I'm going to go into this one more time. Thank you. You're welcome. Thank you.

17 Various QRS Configurations

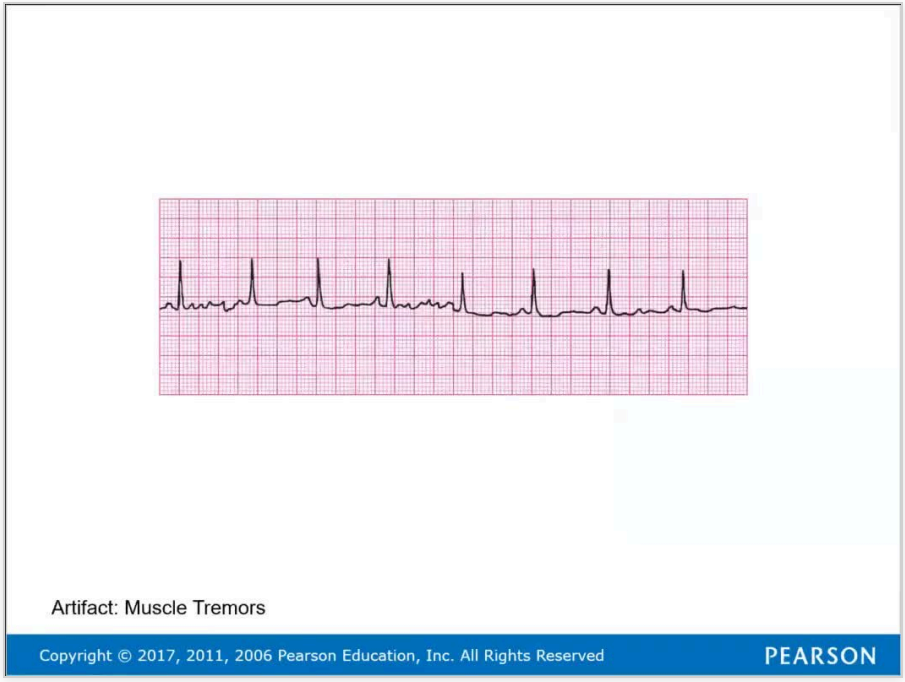
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18 Artifact: Muscle Tremors

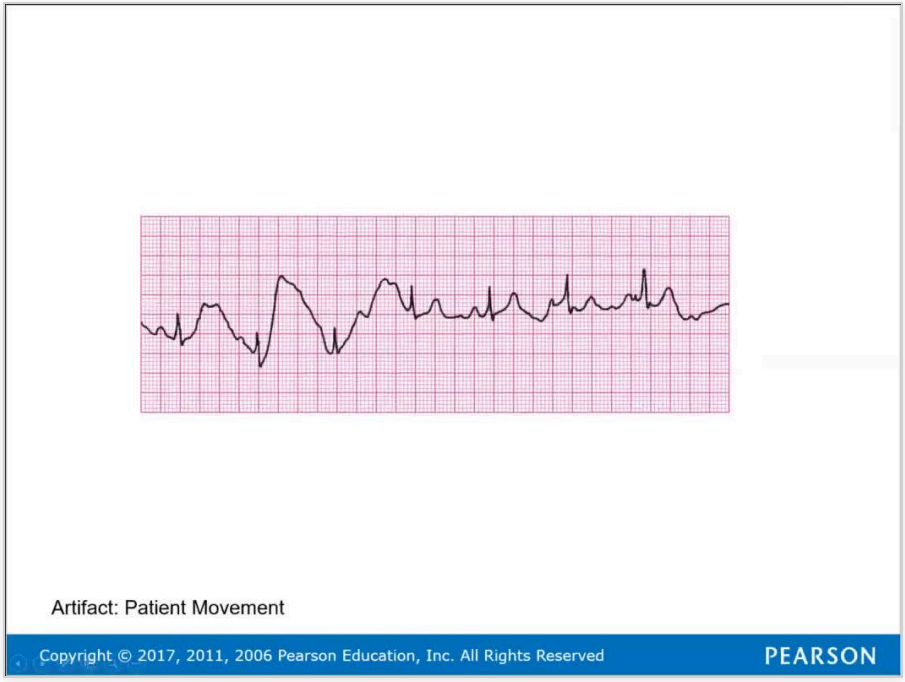
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19 Artifact: Patient Movement

0:32:52 – 0:33:18



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